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| **PI Name:** | Hahn, Inseob Ph.D. | | |
| **Project Title:** | Investigation of equilibration near the liquid-gas critical point in microgravity utilizing DECLIC/ALI facility | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | FUNDAMENTAL PHYSICS | | |
| **Program/Discipline--Element/Subdiscipline:** | FUNDAMENTAL PHYSICS--Fundamental physics | | |
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| **Space Biology Special Category:** | None | | |
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| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA JPL |
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| **Flight Program:** | ISS | | |
| **Flight Assignment:** | ISS | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Barmatz, Martin ( Jet Propulsion Laboratory )<br>Zhong, Fang ( Jet Propulsion Laboratory )<br>Beysens, Daniel ( Commissariat a l'Energie Atomique at Grenoble and Ecole Supérieure de Physique et Chimie Industrielle )<br>Garrabos, Yves ( Centre National de la Recherche Scientifique (CNRS)-)-Institut de Chimie de la Matière Condensee de Bordeaux (ICMCB) Laboratory )<br>Lecoutre, Carole ( Centre National de la Recherche Scientifique (CNRS)-Institut de Chimie de la Matière Condensee de Bordeaux (ICMCB) Laboratory ) | | |
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| **Performance Goal Text:** | | | |

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| **Task Description:** | A key feature of a continuous phase transition in many physical systems in Nature is scale-invariance in fluctuations of the order parameter, which implies continuity between large and small scale fluctuations associated with the transition. Scale-invariance occurs in many systems that are still not completely understood, such as the universe at the onset of the big bang as currently envisioned by cosmologist, the clustering of galaxies, massless fields in elementary particle physics, the distribution of earthquakes, and turbulence in fluids and plasmas. In each of these examples there is a wide range of scales over which some phenomena vary as a power-law of the scale. The Nobel Prize winning, Renormalization Group theory is the most successful approach thus far developed to derive and explore the relations between the characteristics of a system as viewed on different length scales. Many scientific investigations using simple fluid systems have been performed on the ground. The predictions of the RG theory have been reasonably well validated. Nevertheless, the theory is based on “seemingly innocuous” assumptions that will stand or fall based on more sophisticated experiments. Experimental studies of critical phenomena using simple fluids on Earth have been very difficult to perform due to the gravity induced density gradient.<br>NASA’s MISTE/COEX (microgravity scaling theory experiment/coexistence curve experiment) flight experiments were high precision studies of static critical phenomena planned on board the ISS. Those experiments were to address one of the most difficult experimental challenges in studying static properties of near critical fluid in microgravity, which is associated with the equilibration time of the sample. Recently, the DECLIC (Dispositif pour l'Etude de la Croissance et des LIquides Critiques) facility supported by CNES/NASA was successfully integrated into the International Space Station (ISS). The DECLIC facility developed by CNES investigators allows precision studies of non-equilibrium phenomena near the liquid-gas critical point. Multiple science inserts for the DECLIC facility have been developed to study room temperature and high temperature critical fluids. These inserts are currently on-board ISS to study the boiling phenomena, phase separation and piston effect of near critical fluids. The DECLIC/ALI experiment is to be performed in the near future.<br><br>Detailed discussions with the DECLIC PI’s have clearly indicated that additional important scientific investigations can be carried out utilizing the available DECLIC/ALI hardware. Taking advantage of the current CNES funded DECLIC facility, we propose to NASA to study the temperature and density relaxation behavior of the fluid (SF6) near its liquid-gas critical point by utilizing DECLIC/ALI experiments. By analyzing experimental data taken under microgravity conditions, critical data to understand the equilibration processes of a near critical fluid, especially in the two-phase region, can be obtained that is not feasible on the Earth. The results of data analysis, modeling, simulations, and ground support experiments will enhance the scientific return of the proposed DECLIC investigations. This study of equilibration near a critical fluid will directly support the objectives of NASA’s MISTE/COEX flight experiment and benefit future critical phenomena studies utilizing the DECLIC facility.<br><br>The research team of this proposal consists of NASA MISTE/COEX investigators and CNES DECLIC principal investigators. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | This work will provide an unique opportunity to measure experimental properties of a fluid near its liquid-gas critical point without gravitational effects and to refine the theoretical equation of state models. The critical and supercritical state of fluids exhibit unique properties. For instance, the fluid near the critical point and supercritical state shows very large compressibility. The use of these fluids, for instance, for the storage of cryogenic propellant, under normal gravity and/or reduced gravity conditions raises fundamental questions regarding fluid dynamics, heat transfer, interfacial phenomenon. These experimental work will enhance the understanding of basic physics regarding the fluids near critical state. Potential impacts are fields of applications like supercritical fluid extraction, Dry-cleaning, nano and micro particle formation, supercritical drying, supercritical water oxidation. |
| **Task Progress:** | The DECLIC is a muti-user facility to investigate critical fluids on-board ISS. The one of the instrument is the ALI insert to the study of SF6 (sulphur hexafluoride) near its liquid-gas critical point. The ALI commissioning was performed in March 2011 and two experimental sequences have already been performed. These sequences are dedicated for studies of boiling phenomena as originally planned by CNES investigators. Further utilization of the insert to study the critical phenomena have been added to the program. The following scientific objectives are defined for the mid-term agenda of the DECLIC activities.<br>The objectives of the new proposed NASA science investigation are to measure thermal diffusivity, heat capacity at constant volume, turbidity, and coexistence curve in two-phase domain of a fluid sample near the liquid-gas critical point. This investigation will utilize the existing ALI insert as is. After completion of the original scientific objectives of the ALI insert, additional experimental scripts will be added to the DECLIC activities on board ISS.<br><br>To measure the thermal diffusivity and heat capacity, a heat pulse technique will be used using a heater and thermistors in the Interferometric Observation Cell (IOC). By monitoring the sample temperature after the heat pulse, thermal diffusivity and heat capacity at constant volume can be obtained. The thermal diffusivity will be also determined by monitoring evolution of density profile of the sample after the heat pulse using the interferogram. We will also use the Direct Observation Cell (DOC) to measure the turbidity and coexistence curve of the sample. As the average density of the sample is 2.5% higher than the critical density, we will measure the turbidity of the sample at this off-criticality as a function of temperature. The results will be compared with the previous MIR/ALICE 2 results with the sample density at 0.8% off criticality. These turbidity measurements at two different densities in microgravity condition will enlighten the comparison with Ornstein-Zernike theory. Density measurements in the two-phase domain will be also performed by determination of the volume of the gas bubble in the DOC. This is feasible because the bubble is expected in middle of the DOC cell due to its curved window. By measuring the volume of the gas bubble at different reduced temperatures, the coexistence curve of the fluid can be determined. The experimental result will be compared with predictions of the recent crossover equation-of-state theoretical models.<br><br>Recently we have made a progress in comparison between theoretical crossover equation of state model (CPM) and a previous mission data. The previous MIR/ALICE 2 mission to measure the turbidity of SF6 fluid near the critical point. The off-critical state (0.8% off the critical density) data and the calculated turbidity using the CPM model shows good agreement. We plan to measure the turbidity of SF6 using ALI insert in ISS in the near future and its result will be compared with the prediction. We are also making a progress on a manuscript on re-analyis of the compressibility and turbidity measurements using both massive renormalization group crossover equation of state and CPM model. |

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| **Bibliography Type:** | Description: (Last Updated: 01/02/2025) |
| **Papers from Meeting Proceedings** | Pont G, Barde S, Blonde D, Zappoli B, Garrabos Y, Lecoutre C, Beysens D, Hicks M, Hegde U, Hahn I, Bergeon B, Billia B, Mangelinck-Noel N, Ramirez A, Trivedi R. "DECLIC, soon two years of successful operation." Paper presented at 62nd International Astronautical Congress, Cape Town, South Africa, October 3-7, 2011. Session: Facilities and Operations of Microgravity Experiments.<br>62nd International Astronautical Congress, Cape Town, South Africa, October 3-7, 2011. , Oct-2011 |